

sequence

Intro to Alignment Algorithms: Global + Local

- evolution — most integral concept to this class
- evolution models = mathematical models
- how can we use algorithms to find important, non-random similarities between species w/ a common ancestor?

Global Alignment

Sequence Comparison

Biomolecular sequences:

- DNA: A, T, C, G
- RNA: A, U, C, G
- Protein: 20 amino acids

Sequence similarity helps us to discover new genes + proteins

Basic Similarity Analysis Algorithm

Global

- Scoring schemes
- edit graphs — 2 sequences
- alignment = "path in edit graph"
- principle of optimality
- dynamic programming alg.
- trace back

Sequence Alignment

- input: 2 sequences w/ same alphabet [but not necessarily the same length]
- output: alignment
- vocab:
 - match $\begin{smallmatrix} A \\ A \end{smallmatrix}$
 - mismatch $\begin{smallmatrix} A \\ C \end{smallmatrix}$
 - indel $\begin{smallmatrix} A \\ _ \end{smallmatrix}$
- ↳ "insertion or deletion"

Formalizing

$$\begin{aligned} X &= x_1, x_2, \dots, x_n \\ Y &= y_1, y_2, \dots, y_m \end{aligned}$$

} some sequences of lengths n and m

Scoring Schemes

Unit Score

δ	A	C	G	T	-
A	1	0	0	0	0
C	0	1	0	0	0
G	0	0	1	0	0
T	0	0	0	1	0
-	0	0	0	0	1

high score = identical

← example for DNA

Alignment

◦ What is the maximum score / alignment?

A C G
| | |
A G G

$$\circ \text{score} = \sum \delta(x, y) = \delta(A, A) + \delta(C, G) + \delta(G, G) = 1 + 0 + 1 = 2$$

Gaps - you can put them anywhere to maximize alignment

A A A G G G

G G G A A A

score = 0

A A A G G G - - -

- - - G G G A A A

score = 3

ex) A - C G - G

A T C G T G

$$\text{score} = \delta(A, A) + \delta(-, T) + \delta(C, C) + \delta(G, G) + \delta(-, T) + \delta(G, G)$$

Scoring Fxns

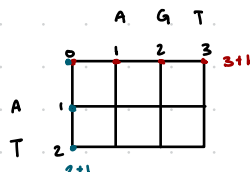
scoring fxn = sum of terms each for a pair of aligned residues / gaps

Edit Graph

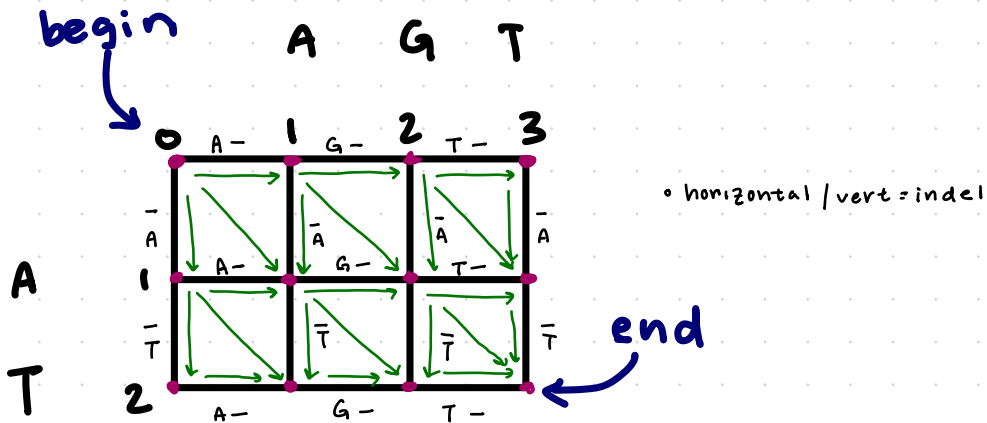
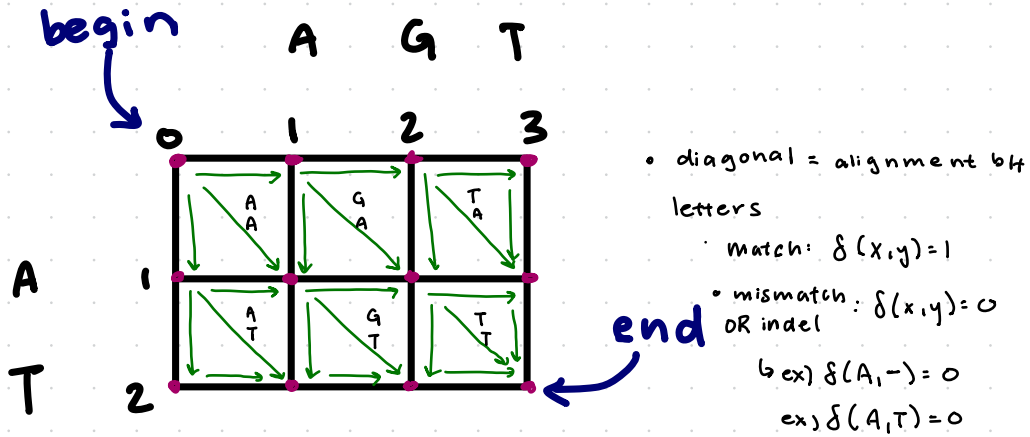
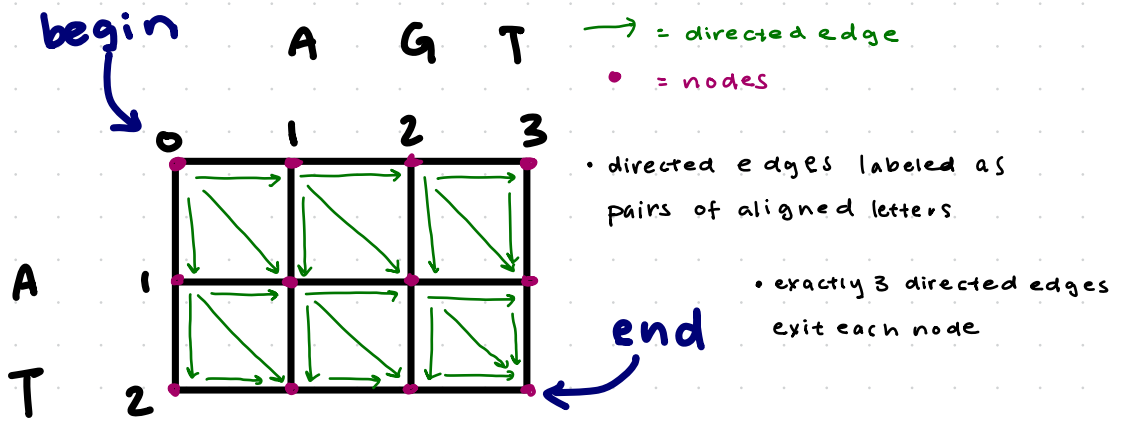
◦ highest scoring alignment?

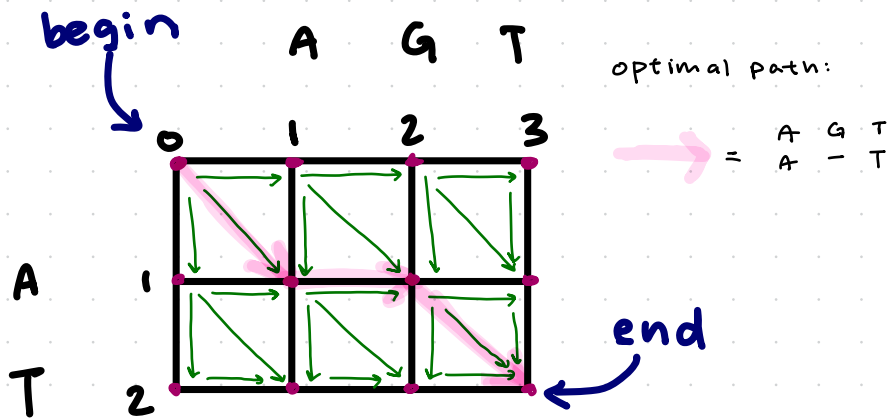
◦ path b/wn begin & end nodes = Edit graph

↳ ex) A G T and A T alignment



Edit graph has $(3+1) * (2+1)$ nodes





Principle of Optimality

The optimal answer to a problem is expressed in terms of optimal answers for its sub-problems

Dynamic Programming

Given 2 sequences X and Y and scoring scheme δ

Find optimal alignment (max score)

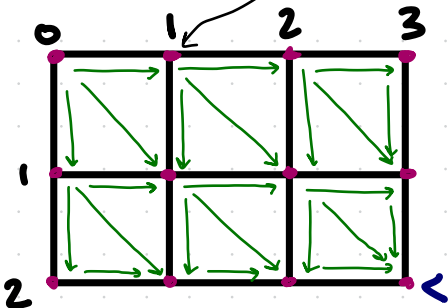
Part 1: compute optimal alignment score

Part 2: construct alignment

↳ "trace back"

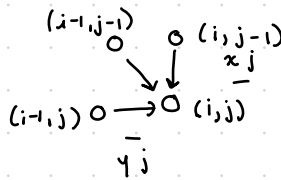
DP Matrix $S(i,j)$

value @ node
 $S(1,0)$



Matrix $S = [S(i,j)]$

$S(i,j)$ = score of max cost path from beginning to end $(0,0) \rightarrow (i,j)$



$S(i,j) = \max$

vertical path:
 $S(i,j-1) + \delta(x_i, -)$

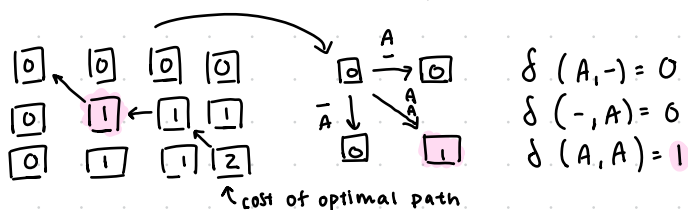
horizontal path:
 $S(i-1,j) + \delta(-, y_j)$

diagonal path:
 $S(i-1,j-1) + \delta(x_i, y_j)$

value of inc. node $\square$ scoring scheme $\rightarrow$

Trace back

- initialize 1st column / row w/ ϕ s



- calculate the cost of each potential path $\delta(x, y)$
- max cost = value of node i
- follow arrows to get optimal alignment

Local Alignment

$$S(i, j) = \max \begin{cases} \phi \leftarrow \text{add } \phi'' \\ S(i-1, j-1) + \delta(x_i, y_j) \\ S(i-1, j) + \delta(-, y_j) \\ S(i, j-1) + \delta(x_i, -) \end{cases}$$

General Scoring Schemes

Assumptions:

- independence of mutations @ diff sites
↳ additive scoring scheme ***
- gaps of any length are considered 1 mutation

Heuristic Interpretation of the "alignment score"

↳ likelihood, statistics

ex) $X = A A C G T A A$ introduced 2 gaps (horizontal lines) [indel]
 $Y = A C G G T G T$ we have 4 matches (vertical lines)
 try: $\begin{array}{ccccccc} A & A & C & G & - & T & A & A \\ | & | & | & | & & | & | & | \\ A & - & C & G & G & T & G & T \end{array}$ we have 2 mismatches ($\begin{smallmatrix} A \\ G \end{smallmatrix}, \begin{smallmatrix} A \\ T \end{smallmatrix}$) [substitution]

Let p be the probability of a match. Let g be the probability of a mismatch.
 Let v be the probability of a single letter indel.

A A C G - T A A
| | | |
A - C G G T G T

The log-likelihood of $P_{\text{alignment}}$ is $S' = \log(P_{\text{alignment}}) = 4 \log(p) + 2 \log(q) + 2 \log(r)$
 $\quad \quad \quad \underline{4 \log k + 2 \log k + 2 \log k}$

$$\mu = \log\left(\frac{k}{g}\right) \text{ mV}$$

choose k such that

$$\log(P/k) = 1$$

$$= 4 \log\left(\frac{p}{K}\right) + 2 \log\left(\frac{q}{K}\right) + 2 \log\left(\frac{r}{K}\right)$$

penalties - μ = mismatch

$$= 4 - 2\mu - 2\gamma \quad \gamma = \text{gamma}$$

score of alignment